

SIH26091 | GRAMARTHA

Deep Engine and Product Completion Report

Version 0.5.0 | Audited 28 August 2026

West Bengal hyper-local venture planning, evidence freshness, trained models, finite optimization, joint uncertainty, robust choice, finance, and verified product behavior.

VERIFIED	BOUNDARY
Local HTTP product, 23 district smoke checks, 7 deep E2E cases, 50 tests, trained artifacts.	Conditional benchmark, not observed 2026 locality market or lender approval.

This report records the strongest honestly verified product at this checkpoint. It distinguishes observed data, historical baselines, projections, modelled benchmarks, and unknown current values. It does not claim an observed 2026 locality market, lender approval, guaranteed income, or a global optimum outside the enumerated venture configurations.

1. Executive outcome

GramArtha is now a working local web product and an evidence-aware analytical engine for West Bengal entrepreneurs. A user selects one of 23 database-supported district labels, searches only within that district, chooses a locality and catchment, optionally supplies an entrepreneur profile, and requests either a quick or deep analysis. The ordinary path can compare ten sector adapters, generate finite starter/growth venture configurations, calculate 36-month cash behavior, and explain opportunities, risks, financing, and next actions.

Deep mode adds 512 reproducible joint scenarios, survival and payback frequencies, downside cash, VaR/CVaR loss, Pareto filtering, exact regret over the simulated candidate table, numerical failure boundaries, and separate expected-value, survival-first, CVaR-aware, and minimax-regret winners. The selected output remains **CONDITIONAL** because its market envelope is a low-confidence benchmark derived from regional ASUSE priors rather than observed locality transactions.

Verified headline status:

- 50 automated Python tests pass; Ruff and JavaScript syntax checks pass.
- 23/23 district requests return HTTP 200; 22 produce conditional benchmark plans and one is safely gated.
- Seven diverse deep E2E cases run with 512 scenarios each.
- A real Chromium session completed the Kolkata best-opportunity flow with no console warning or error.
- The current public Netlify URL is not this local product and is not claimed as deployed.

2. User problem and decision contract

The product is designed for an ordinary entrepreneur who asks: what small venture might fit this place, my capital, and my risk tolerance; what would it cost; what could go wrong; and what should I verify before spending money? It is not a survey explorer or a raw model dashboard.

Every output follows an explicit contract. Historical evidence keeps its observation year. Modelled values are labelled MODELLED_BENCHMARK. Projected population retains base year, rate scenario, horizon, and uncertainty. Missing current evidence produces a caveat or evidence gate. Finance schemes are screens, not sanctions. Candidate selection is exact only over the configurations actually generated.

The response therefore separates four questions:

- Is the location identity and catchment technically resolved?
- Is there enough evidence to construct a planning benchmark?
- Which enumerated plan is most robust under stated shocks?
- What current facts must a person verify before committing money?

3. Product journey

The implemented browser journey is:

1. Area: West Bengal is fixed; district is loaded from the evidence database. 2. Locality: type-ahead is district-scoped and returns stable geographic identities. 3. Catchment: user selects 3, 5, 10, 15, or 20 km. 4. Profile: capital is required; income goal, debt ceiling, experience, skills, assets, and risk tolerance are optional. 5. Opportunity: choose a sector or ask GramArtha to compare the supported library. 6. Analysis: quick mode computes deterministic planning outputs; deep mode adds 512 uncertainty scenarios. 7. Decision: tabs present summary, local market, opportunities, risk and SWOT, business plan, finance, action plan, and technical evidence.

The UI intentionally leads with the business decision. Internal IDs, evidence-gate codes, formulas, and raw evidence remain available in a secondary disclosure instead of dominating the customer experience.

4. Verified statewide geographic coverage

Layer	Verified extent	Interpretation
Geographic identity store	53,537 identities	West Bengal village, town, and ward identities from integrated official/proxy sources.
Census location directory	41,154 WB rows	Official 2011 location codes; structural/historical use only.
Census PCA evidence	179,148 records	Population, household and sex fields across 19 districts as defined in 2011.
Livestock identities	40,475 localities	Publisher labels across 23 district groups.
Livestock species records	202,375	Historical 2019 stock context, not current productive supply.
OSM road ways	633,601	Statewide routing proxy.
OSM POIs/places	17,212	Volunteered facility/place proxy with completeness caveat.
OSM admin areas	381	Administrative proxy polygons/relations where present.
Survey district groups	23	HCES/ASUSE sampled regional priors, not exact locality observations.

Statewide geographic coverage does not imply statewide economic completeness. District names also span historical and current publisher vocabularies. The all-district smoke test deliberately leaves undivided 2011 Bardhaman gated when a safe mapping to the current ASUSE prior cannot be established.

5. District-scoped locality resolution

The district dropdown is generated from the database rather than hard-coded in the browser. Search accepts a query, limit, and district filter. The evidence store normalizes the Bardhaman/Barddhaman spelling family for discovery while retaining distinct publisher identities where split-district semantics matter.

This design prevents a common failure: choosing a locality with the right name in the wrong district. The selected response returns stable geo_id, name, district, locality type, latitude/longitude where available, and evidence provenance. Search is a resolution layer, not proof that every identity has current economic evidence.

The 23-label smoke run produced HTTP 200 for every district. Twenty-two sample localities produced a conditional finite-set venture. Historical Barddhaman returned INSUFFICIENT_EVIDENCE with NO_DEMAND_EVIDENCE rather than borrowing a potentially wrong current district prior.

6. Data inventory and years

Dataset	Observation/version	Freshness class	Decision use
Census geography/PCA	2011	HISTORICAL_BASELINE	Stable identity, settlement type, historical population anchor.
20th Livestock Census	2019	STALE_FOR_DECISION	Historical species stock context only.
HCES comparison wave	2022-23	STALE_FOR_DECISION	Historical comparison.
HCES production wave	Aug 2023-Jul 2024	RECENT	District/sector consumption rate prior.
ASUSE comparison wave	Oct 2023-Sep 2024	historical comparison	Aggregate enterprise comparison.
ASUSE production wave	Jan-Dec 2025	RECENT	District/sector/NIC2 enterprise benchmark.
BAHS	2025 publication	RECENT state context	State-level yield context, not locality supply.
OSM extract	retrieved 2026-08-27	CURRENT structural proxy	Roads, catchments and POI/place proxies.
PMMY official page	updated 2026-02-05	CURRENT rule screen	Category/purpose/collateral screening only.
AHIDF temporary continuation	2026-04-01 to 2026-09-30	CURRENT rule screen	Conditional scheme screen subject to superseding approval.

Unknown current locality variables include observed selling price, procurement price, rent, wage, electricity/fuel logistics cost, verified competitor capacity, supplier quotes, and lender-specific rate/underwriting. They are never silently replaced with an old observation.

7. Restricted source integrity

The four official survey archives were acquired under applicant access, retained privately, and excluded from public-share packaging. Full member CRC checks passed.

Archive	Exact bytes	SHA-256
HCES 2022-23 CSV	244,317,752	093a51d337eb07eede7d6e2a00ec55790d926bacf869be2cf2825abd629444ae
HCES 2023-24 CSV	256,315,433	acf7b9cc840676fb812c05c48f09fa034955fe2cad5112e6d9b6d852f5f2e267
ASUSE 2023-24 CSV	98,219,217	82a2e59aab71fee6dda7cd5f859c7cfbcf5e23ddf70b36a5a2962296b838de54
ASUSE calendar-2025 CSV	173,738,167	f20b35abdb7ba97daaca24f4b927a277e95e1145fcb86b2bd97304f771dc591

The aggregate priors, public-safe metrics, provenance, and code are shareable. Unit respondent records and fitted artifacts derived from restricted records remain private. This is a legal/data-governance boundary rather than a technical omission.

8. Freshness audit and evidence statuses

Every important evidence record can retain source, dataset/version, observation date, effective date, retrieval date, geography, units, estimate status, confidence, and freshness status. The supported freshness labels are CURRENT, RECENT, HISTORICAL_BASELINE, PROJECTED, STALE_FOR_DECISION, and UNKNOWN.

Structural variables can legitimately use an older authoritative baseline when the year is visible. Fast-changing variables must use a current source or remain uncertain. A recommendation may use a modelled planning range, but it cannot call that range an observed current locality value.

The audit policy is fail-closed for commodity-specific dairy decisions: current productive supply, price, incumbent capacity, route cost, and source-linked cost configuration must exist. Generic sector adapters are explicitly a lower-confidence planning path based on the most recent available ASUSE regional enterprise prior.

9. Population projection policy

Census 2011 is never relabelled as 2026. Where a generic calculation requires a population scale, the engine can create an explicit scenario projection with base year 2011 and horizon 15 years:

- Rural annual rates: 0.6 percent lower, 1.1 percent central, 1.6 percent upper.
- Urban annual rates: 1.0 percent lower, 1.7 percent central, 2.4 percent upper.
- Formula: $P_{2026} = P_{2011} * (1 + g)^{15}$.
- Status: PROJECTED; confidence: LOW.

The projection notes that boundary-change adjustment is unavailable. These scenario rates are planning assumptions, not official locality forecasts. The older v0.3 report correctly said no projection was stored at that checkpoint; v0.5 adds the explicit method and preserves the change in versioned documentation.

10. HCES model training

The HCES task predicts zero-inclusive monthly liquid-milk litres per household member. Training used 18,120 West Bengal household rows spanning 23 district groups; 7,560 targets were zero. Validation used leave-one-district-out geographic holdout.

Model	MAE litres	RMSE litres	Result
Weighted category-mean baseline	1.384585	1.748752	Baseline
Ridge one-hot	1.367413	1.733946	Improved
Random forest	1.353607	1.724402	Holdout winner

Random-forest weighted bias was 0.000133, mean calibration ratio 1.000082, and decile calibration MAE 0.131882. The fitted artifact is real and checksum-registered. Ordinary locality requests do not contain the household microfeatures required for valid inference, so production uses the direct weighted district/sector HCES prior instead of pretending to score an inapplicable row.

11. ASUSE model training

The ASUSE task predicts annualized enterprise gross value added in INR using official item 769 and the documented reference-period rule. Training used 38,626 West Bengal enterprise rows across 23 district groups; 40 targets were zero. Validation used leave-one-district-out holdout.

Model	MAE INR	RMSE INR	Result
Weighted category-mean baseline	97,124.15	259,004.56	Baseline
Ridge one-hot	68,004.76	197,753.34	Improved
Random forest	61,573.20	193,228.67	Holdout winner

Random-forest weighted bias was INR 22.54, mean calibration ratio 1.000151, and decile calibration MAE INR 2,528.68. Production again uses the direct weighted district/sector/NIC2 prior because the customer request lacks enterprise microfeatures. Training completion and production applicability are reported separately.

12. Sector adapter library

The generic planning library contains ten supported adapters mapped to the latest available ASUSE production priors:

Adapter	NIC2 basis	Planning role
Kirana/grocery	47	Small retail benchmark
Poultry input/egg aggregation	46	Wholesale/distribution benchmark
Fish collection/distribution	46	Cold-chain-light distribution benchmark
General food processing	10	Small manufacturing benchmark
Rural distribution/aggregation	46	Local logistics/wholesale benchmark
Flour mill	10	Staple processing benchmark
Spice processing	10	Value-add processing benchmark
Mustard oil extraction	10	Oilseed processing benchmark
Electronics/mobile retail	47	Durable retail benchmark
Household-goods distribution	46	Wholesale benchmark

Each adapter declares unit, ASUSE mapping, demand/incumbent scaling, starter/growth capacities, CAPEX/working-capital split, fixed overhead, staffing, space, service radius, inventory/receivable/payable days, lifetime, residual value, licensing assumptions, and downside variables. Dairy remains the deeper physical-evidence path and is gated rather than approximated by a generic adapter.

13. Demand, supply, price and competition ladder

The estimation ladder prefers direct current locality evidence, then recent regional official priors, then explicit projections or modelled benchmarks, and finally a gate. It never silently treats a lower rung as a higher rung.

Generic demand is an ASUSE-scaled planning envelope, not exact household demand. Supply and incumbent service are benchmark envelopes, not verified competitor capacity. The browser displays ranges and confidence, not false point precision. OSM POIs can contextualize nearby activity but are not a complete business census.

Commodity-specific dairy demand may combine an HCES rate with an explicit projected population range. Physical supply requires current productive animal share, yield, seasonality, collection loss, and reachable capacity. Current selling/procurement price and route cost remain mandatory for a high-confidence dairy decision.

14. Catchment, graph and flow mathematics

The spatial layer resolves a catchment against the West Bengal OSM extract. A central Kolkata 5 km test returned 1,172 entities; a tested local route was 4.63 km. The graph represents sources, incumbent/venture transformations, transport links, and demand sinks when decision-ready evidence exists.

The flow objective is lexicographic:

- First maximize served demand F .
- Then minimize total economic cost $\sum_e (c_e * f_e)$ subject to the maximum F .
- Constraints: $0 \leq f_e \leq u_e$ and flow conservation at transshipment nodes.

Bottlenecks are ranked by marginal served-demand gain after an edge-capacity perturbation. Counterfactual output separates baseline served demand, newly served demand, venture flow, and incumbent flow displaced by the new venture. This prevents cannibalized flow from being mislabelled as newly created welfare.

15. Venture primitives and cash conversion

A venture primitive now carries operational structure, not merely cost and capacity:

- service radius and required space;
- operating days per month;
- inventory, receivable, and payable days;
- cash conversion cycle = inventory days + receivable days - payable days;
- lifetime and residual value;
- staff, CAPEX, working capital, overhead, and licensing assumptions.

The customer view exposes the practical meaning: premises, staff, licenses, stock cycle, service radius, startup cost, monthly operating burden, and the evidence to verify. Financial calculations preserve the distinction between startup investment, variable procurement, fixed operating cost, and financing.

16. MVV and finite-set optimization

For each adapter, the engine enumerates feasible starter/growth configurations. Minimum viable venture selection is exact over that finite list. A candidate must respect capital and any supplied income/debt constraints; the engine does not claim a continuous global optimum or a complete mixed-integer model of all possible business designs.

The opportunity table distinguishes useful roles:

- Lowest viable: minimum feasible project cost.
- Highest upside: strongest central cumulative cash result.
- Survival-first: highest scenario survival, then downside tie-breaks.
- Robust: minimax-regret winner over the explicit scenario table.

The Pareto frontier retains candidates not dominated on lower investment, higher survival, and higher median cumulative cash. This prevents one arbitrary weighted score from hiding tradeoffs.

17. Joint uncertainty model

Deep mode draws 512 reproducible scenarios from seeded triangular distributions. The seed is derived with SHA-256 from stable analysis/candidate identity inputs. Shocks cover demand, selling-price/revenue factor, variable cost, and fixed cost. Triangular distributions are used because they make lower, modal, and upper planning assumptions explicit and avoid an unjustified Gaussian claim.

For each candidate the engine records:

- survival rate over 36 months;
- target-income attainment rate when a goal is supplied;
- payback-within-36-month rate;
- p10 minimum cash and p10 cumulative cash;
- median cumulative cash;
- 95 percent value-at-risk loss and conditional value-at-risk loss.

Scenario survival is model survival under the stated shock envelope. It is not the empirical probability that a real business will survive.

18. Robust choice and regret

Let $C_s(c)$ be cumulative cash for candidate c in scenario s . Scenario regret is $\max_j C_s(j) - C_s(c)$. Maximum regret is the worst scenario regret for a candidate. The minimax-regret choice is $\operatorname{argmin}_c \max_s \operatorname{regret}_s(c)$.

Expected-value, survival-first, CVaR-aware, and minimax-regret choices are computed separately. They may disagree, and the product exposes that disagreement instead of forcing one answer. The selected ordinary recommendation prioritizes scenario survival and downside cash before central upside where the user has not supplied a different explicit objective.

CVaR is presented as worst-tail cumulative cash, not as a confusing negative loss sign. The finance screen also distinguishes operating break-even, cash break-even, and investment payback.

19. Failure boundaries and sensitivities

The engine uses a 40-iteration numerical search to find boundaries for demand factor, selling-price/revenue factor, and variable-cost factor. If the entire tested range survives, the output says that no cash failure occurred within the tested 36-month range. It does not manufacture a near-zero threshold.

These boundaries answer questions such as: how far can demand fall before monthly cash becomes negative; how far can price/revenue fall; how far can variable cost rise? The browser pairs each boundary with scenario p10 cash, scenario survival, payback frequency, and tail cash so the user can see both one-variable and joint-shock behavior.

The current engine does not estimate an empirical correlation matrix from local time-series data. Scenario factors are planning assumptions. Therefore sensitivity findings are useful for stress design, not calibrated real-world probabilities.

20. Finance and accounting

The engine calculates project cost, own capital applied, external finance gap, monthly revenue/cost/cash, gross margin, operating margin, contribution margin per planning unit, break-even volume, payback, 36-month NPV, and annualized IRR.

Key formulas:

- Break-even volume = fixed cost / contribution margin per unit.
- $NPV = \sum_t CF_t / (1 + r_{\text{monthly}})^t$, with $r_{\text{monthly}} = (1 + r_{\text{annual}})^{(1/12)} - 1$.
- IRR solves $NPV(r) = 0$ by bounded bisection and is annualized.
- Payback month is the first month cumulative venture cash recovers startup investment.

The displayed NPV uses a 12 percent annual planning discount rate. IRR and NPV are benchmark-adjusted outputs, not guaranteed investment returns. Input/output accounting avoids double-counting ASUSE procurement inputs in both margin and fixed OPEX.

21. Finance rules and effective dates

PMMY screening uses the Department of Financial Services page updated 2026-02-05. Categories are Shishu up to INR 50,000; Kishore above INR 50,000 through INR 5 lakh; Tarun above INR 5 lakh through INR 10 lakh; and Tarun Plus above INR 10 lakh through INR 20 lakh for a borrower who previously took and successfully repaid Tarun. The official rule does not establish one universal lender rate, tenure, underwriting result, or sanction.

AHIDF/IDF is temporarily continued from 2026-04-01 through 2026-09-30 or until an earlier superseding approval. Existing temporary-period terms include 3 percent interest subvention, lender finance up to 90 percent of eligible project cost, and up to eight years including up to two years of principal moratorium. Portal window, eligible cost, margin, gross/net lender rate, security, underwriting, and sanction must be confirmed live.

22. Computed SWOT and practical advice

SWOT is assembled from calculation outputs rather than generic template text. Strengths can include low financing dependence, positive contribution margin, or robust survival under the tested envelope. Weaknesses can include low confidence, long cash conversion, thin margin, or reliance on a regional benchmark. Opportunities can include an unserved modelled envelope or a lower-capital Pareto option. Threats can include downside failure boundaries, stale physical supply, unknown current prices, or scheme expiry.

The Action Plan tab converts these findings into stages: verify local demand and current prices; obtain supplier and premises quotations; validate licenses; run a small pilot; measure weekly revenue, margin, inventory days and repeat demand; expand only after stated triggers. A separate When Not To Start section identifies evidence or cash conditions that invalidate the plan.

23. Deep E2E validation

Locality / district	Sector	Capital INR	Cost INR	Payback	Scenarios	Result
Kolkata / Kolkata	Kirana	100,000	35,971	21 mo	512	CONDITIONAL
Barasat / North 24 Parganas	Poultry aggregation	100,000	13,823	21 mo	512	CONDITIONAL
Abad Bhagabanpur / South 24 Parganas	Fish distribution	125,000	57,047	32 mo	512	CONDITIONAL
Kharibari / Darjeeling	Food processing	150,000	54,986	24 mo	512	CONDITIONAL
Anandapur / Jalpaiguri	Rural distribution	120,000	14,027	15 mo	512	CONDITIONAL
Adina / Maldah	Kirana	80,000	20,816	32 mo	512	CONDITIONAL
Adra / Purulia	Poultry aggregation	90,000	49,500	26 mo	512	CONDITIONAL

All seven returned HTTP 200 using methodology decision-v5. All reported modelled survival 1.0 under the configured envelope because starting capital left a material cash buffer; this is not interpreted as observed certainty. In every case the minimax-regret winner was the growth configuration while the ordinary finite MVV selected the starter configuration.

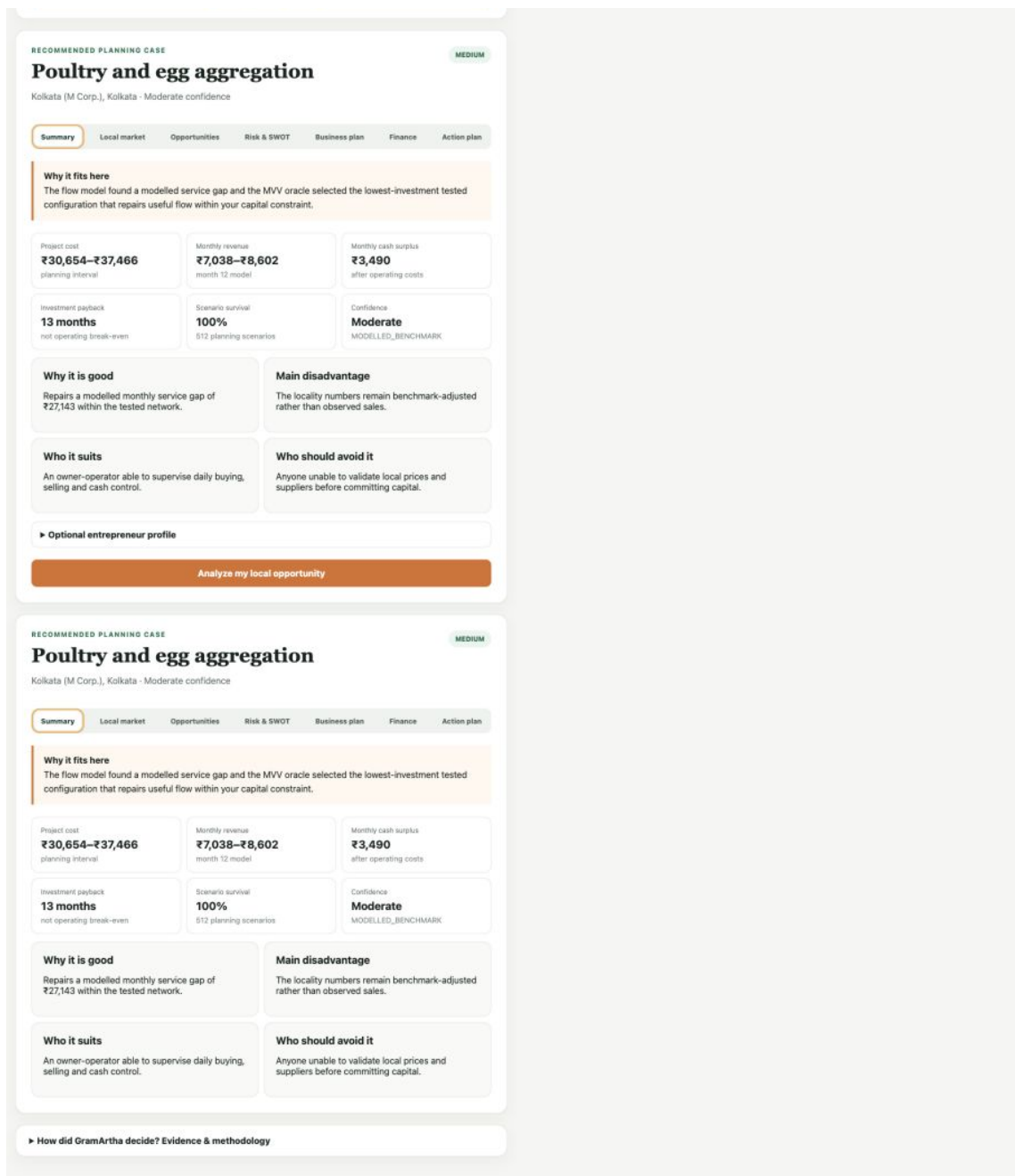
24. All-district smoke test

One database-supported locality was tested for every one of the 23 district labels. HTTP status was 200 in all 23 cases. Twenty-two produced a selected conditional kirana benchmark. The historical undivided Barddhaman case returned INSUFFICIENT_EVIDENCE and NO_DEMAND_EVIDENCE.

The smoke run covers Alipurduar, Bankura, Barddhaman, Birbhum, Dakshin Dinajpur, Darjiling, Haora, Hugli, Jalpaiguri, Jhargram, Kalimpong, Koch Bihar, Kolkata, Maldah, Murshidabad, Nadia, North Twenty Four Parganas, Paschim Barddhaman, Paschim Medinipur, Purba Medinipur, Puruliya, South Twenty Four Parganas, and Uttar Dinajpur.

This is endpoint and evidence-routing coverage, not an economic validation of every district or every locality.

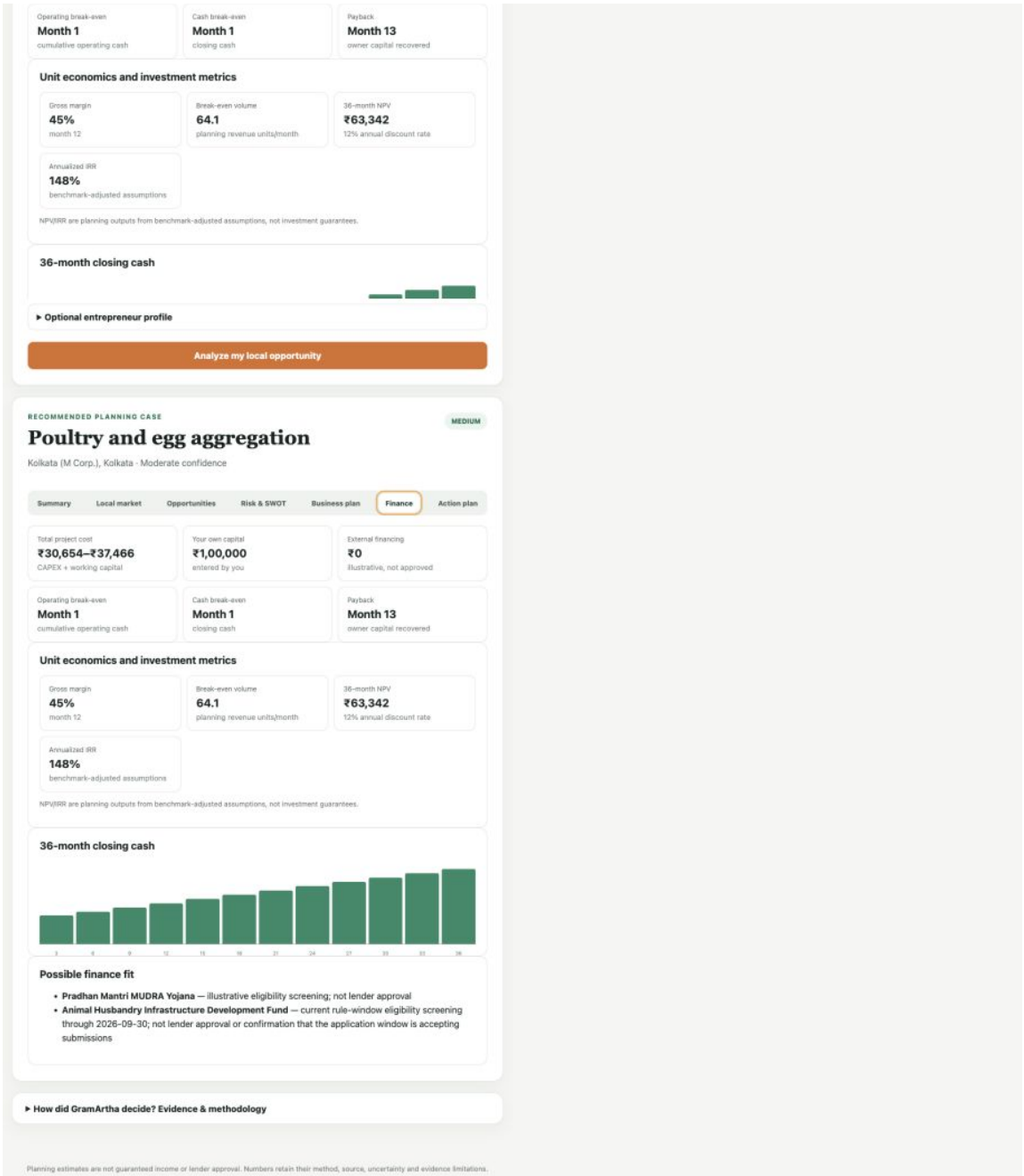
25. Browser validation and product screenshots



The local product was exercised over HTTP in a real Chromium browser. The test loaded exactly 23 district choices, selected Kolkata, ran Find Best across ten adapters in deep mode, completed in about three seconds on the test machine, switched result tabs, and inspected the Finance view. There were zero console warnings/errors.

The screenshot is evidence of rendered software behavior, not proof that the underlying benchmark is an observed market. Public Netlify deployment was not changed and remains outside the verified v0.5 completion claim.

26. Finance screen validation



The browser Finance tab displayed project cost, own capital, financing gap, three break-even concepts, gross margin, break-even volume, NPV, IRR, cash chart, and current-scheme caveats. The Kolkata best-opportunity browser run selected poultry and egg aggregation and displayed a conditional benchmark, not a lender quote.

The UI labels NPV at a 12 percent planning discount and IRR as benchmark-adjusted. It does not infer a loan interest rate when none is available.

27. Verification matrix

Verification	Current result
Python tests	50 passed
Ruff	Passed
JavaScript syntax	Passed
District smoke	23/23 HTTP 200; 22 conditional, 1 gated
Deep product E2E	7/7 HTTP 200 with 512 scenarios
SQLite integrity	Previously audited ok
Registry integrity	53/53 recorded files previously matched size and SHA-256
Restricted ZIP CRC	4/4 passed
Trained artifacts	Two random-forest winners plus ridge artifacts checksum-registered
Accounting identity	1,551 ASUSE groups; max residual INR 3.725e-9
Browser console	Zero warnings/errors in verified flow

Flow benchmark results on chain graphs were 0.00007 s at 10 nodes, 0.00019 s at 50, 0.00210 s at 100, 0.00158 s at 500, and 0.00312 s at 1,000. Bottleneck perturbation was benchmarked through 100 nodes; larger reruns were intentionally omitted because the chain test is not a worst-case graph.

28. Requirement traceability

Requirement family	v0.5 status
Fixed West Bengal, DB districts, scoped search	Implemented and browser-tested
Optional entrepreneur profile	Implemented
Multi-sector library	Ten generic adapters plus gated dairy
Freshness labels and explicit projections	Implemented
Actual model training and holdout metrics	Implemented; production applicability constrained
Catchment, graph, flow, bottleneck	Implemented when decision-ready evidence exists
Automatic venture generation and finite MVV	Implemented for enumerated configurations
Joint scenarios, Pareto, CVaR, regret	Implemented in deep mode
Failure boundaries	Implemented for demand, price/revenue and variable cost
Computed SWOT and action plan	Implemented
Statewide smoke and diverse E2E	Implemented with stated limits
Public deployment	Not completed or claimed
Current locality transactions/quotes	Not acquired; remains an evidence limit

29. Remaining evidence and method limits

- Generic sector results are MODELLED_BENCHMARK, not observed locality demand, sales, competitors, or prices.
- HCES and ASUSE are sampled regional priors and cannot identify exact village demand or enterprise capacity.
- Census 2011 is a historical structural baseline; the 2026 population scenario is a low-confidence explicit projection.
- Current local price, procurement, rent, wage, electricity/fuel, supplier, and competitor-capacity evidence remains unknown.
- Scenario distributions and bounds are planning assumptions, not empirically calibrated survival probabilities.
- MVV and regret are exact only over the generated finite candidate list.
- The automatic library is not a complete universe of ventures and does not include a general configuration MILP.
- OSM is a volunteered spatial proxy and may omit roads, places, or businesses.
- Finance screens do not promise lender rate, eligibility, portal acceptance, sanction, or subsidy receipt.
- Restricted survey unit records and fitted artifacts cannot be included in a public package.
- The public URL presently shows a different site; only the local HTTP product is verified here.

30. Final decision and handoff

v0.5.0 is the strongest honest working product at this checkpoint: a usable local advisory interface backed by real West Bengal identity/spatial data, recent official survey priors, real trained model artifacts and geographic holdout metrics, a finite venture engine, robust joint-scenario analysis, finance/accounting calculations, and broad technical validation.

It is ready for local demonstrations and structured pilot discovery. It is not ready to be described as a high-confidence current-market recommendation service without integrating verified locality transactions, supplier/competitor capacity, current quotes, and lender terms. A safe pilot should collect those fields, compare realized results against the benchmark intervals, recalibrate scenario distributions, and preserve the existing versioned evidence/provenance chain.

The versioned package contains code, configuration, databases, documentation, public-safe outputs, tests, E2E records, screenshots, this report, and a SHA-256 manifest. Restricted respondent data remains outside the shareable archive by design.